

# Yukawa screening derivation of the bond-valence rule

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The bond-valence model is a standard way to estimate bond strengths in crystals, but its exponential dependence on bond length has lacked a derivation from a specific physical interaction. We show that this form emerges as the leading-order limit of screened Coulomb electrostatics and that the fitted bond-valence softness can be interpreted in terms of an electronic screening length. This turns bond valence from an empirical fitting rule into a transferable descriptor of local screened charge response across coordination environments. The resulting theory predicts how the bond-valence parameters should vary with ionic charge and coordination number, and that prediction agrees with 150 fitted valences from 94 cation–oxygen species, including 68 in fourfold coordination and 82 in sixfold coordination, at an abundance-weighted coefficient of determination of 0.986. A comparison with first-principles charge densities shows that the bond-valence shell radius tracks the electronic screening cloud with coefficients of determination of 0.9998 for ten alkali and alkaline-earth oxides and 0.967 for 21 other binary oxides whose nearest-neighbor environments match the theory’s assumptions. The widely used bond-valence model is thus the leading-order expression of screened electrostatics in ionic solids.

Pauling’s observation that the strength of a bond reaching a coordinating anion approximates the cation charge divided by the coordination number [1] has been an organizing principle of crystal chemistry for nearly a century. Brown and Altermatt [2] cast this idea as the bond-valence model: the empirical bond strength between a cation  $i$  and anion  $j$  separated by  $R_{ij}$  is

$$S_{ij} = \exp\left(\frac{R_0 - R_{ij}}{B}\right), \quad (1)$$

where  $R_0$  and  $B$  are parameters fitted to crystallographic data and the valence-sum rule  $\sum_j S_{ij} = z_i$  (formal charge) is enforced. The model is used routinely for structure validation [3–5].

Several lines of work have sought a physical foundation for the bond-valence exponential. Preiser *et al.* [6] interpreted bond valence as localized electrostatic flux constrained by Gauss’s-law. Brown developed this picture further within a broader electrostatic framework [3, 4]. However, in those treatments the exponential distance dependence is postulated rather than derived from a specific potential, leaving  $B$  as an empirical fit parameter with length units but no associated physical grounding. Adams related the softness  $B$  to Morse-potential curvature and motivated non-universal  $B$  values [7, 8], but does not connect  $B$  to a screening length. Modern first-principles work can compute screened local interactions from electronic structure [9, 10], but does not derive the bond-valence form itself. Here we show that a Yukawa expansion obtains the exponential as the leading term of a specific screened potential, ties  $B$  to a measurable screening length, predicts the  $B$ – $R_0$  correlation with no fitted constants, and connects the fitted parameters to electronic structure.

*Screened-flux derivation.* A cation with charge  $Q_i = z_i e$  emits electrostatic flux  $Q_i/\epsilon_0$ . In a Yukawa field with

screening length  $\lambda$ , the signed flux amplitude captured by bond  $i$ – $j$  is  $\tilde{\Phi}_{ij} \propto e^{-R_{ij}/\lambda}(1 + R_{ij}/\lambda)$ . Bond valence is a nonnegative partition of the cation valence among its anion neighbors, and two conditions fix that partition. The valence-sum rule  $\sum_j S_{ij} = z_i$  sets the total. In the unscreened electrostatic picture of Preiser *et al.* [6], flux is partitioned according to Gauss’s law, in which each bond carries a valence proportional to the electrostatic flux it subtends. We retain the same correspondence, but for a the screened field. Screening attenuates the flux, and the sum rule renormalizes the total back to the formal charge, so the screened field determines only the relative apportionment among bonds. Preserving that apportionment means that ratios of bond valences equal ratios of the captured screened-flux magnitudes,

$$\frac{S_{ij}}{S_{ik}} = \frac{w_{ij}}{w_{ik}}, \quad w_{ij} := |\tilde{\Phi}_{ij}|.$$

This guarantees the non-negativity that bond valence requires, imports the distance dependence of the screened interaction into the partition, and introduces a single screening parameter,  $\lambda$ , to scale the distance dependence. Together the two conditions uniquely give

$$S_{ij} = z_i \frac{w_{ij}}{\sum_k w_{ik}}. \quad (2)$$

To linearize, expand  $\ln w_{ij}$  about a reference bond length  $R^*$  chosen within the coordination shell, giving

$$\ln w_{ij} = C_i - \frac{R^*}{\lambda(\lambda + R^*)} (R_{ij} - R^*) + \delta_{ij}, \quad (3)$$

where  $C_i$  is independent of bond index  $j$  and  $\delta_{ij} = O[(R_{ij} - R^*)^2/(\lambda + R^*)^2]$ . Exponentiating the linear term and renormalizing therefore reproduces Eq. (1) at  $R_{ij} = R^*$  with only quadratic log-weight error away from

$R^*$ . Define the leading-order Yukawa exponent denominator

$$B = \frac{\lambda(\lambda + R^*)}{R^*}. \quad (4)$$

This  $B$  is the first-order exponent denominator produced by the Taylor expansion of the Yukawa weight. The screening length  $\lambda$  is the primary physical variable, and the sign of  $\lambda$  distinguishes screening from anti-screening. For  $\lambda > 0$  the Yukawa weight decays with bond length, so flux reaching the more distant anions is attenuated and shorter bonds capture the larger share of the valence. Eq. (4) then gives  $B > 0$ , coinciding with the conventional bond-valence parameter in Eq. (1). For  $-R^* < \lambda < 0$  the weight instead grows with bond length, with the result that the medium enhances the far field rather than attenuating it. Longer bonds carry the larger share of the valence, and Eq. (4) gives  $B < 0$ . This anti-screening branch is realized only rarely. Just two of the 103 fitted species below fall on it. The branch sign of  $B$  is distinct from the sign reversal of the slope across the pole in Eq. (5), which reverses the sense of the  $B$ - $R_0$  correlation between  $z < n$  and  $z > n$  while  $B$  itself remains positive for nearly all species on both sides.

*Parameter-free slope.* In the uniform-bond limit, where  $R_{ij} = \bar{R}$  for all  $j$ , the valence-sum rule gives  $R_0 = \bar{R} + B \ln(z/n)$ . Differentiating:

$$\beta := \frac{dB}{dR_0} = \frac{1}{\ln(z/n)}. \quad (5)$$

This is parameter-free: the slope of the  $B$ - $R_0$  correlation at fixed coordination number  $n$  is determined entirely by the charge-to-coordination ratio  $z/n$ , with a pole at  $z = n$ . Figure 1 shows that Eq. (5) reproduces the fitted slopes for 94 cation-oxygen species at  $n = 4$  and  $n = 6$  (68 at  $n = 4$ , 82 at  $n = 6$ , 150 slopes in total). The weighted  $R^2$  is 0.986, with mean absolute error 0.15 under the weighting scheme defined in the Supplemental Material [11]. Each fitted slope is the regression slope of one species' per-structure ( $R_0, B$ ) fits at fixed  $n$ , constructed as described in the empirical validation below.

*Screening collapse at  $z = n$ .* The pole has direct physical content, but stating it requires a distinction that the valence-sum rule alone does not make. At  $z = n$  the sum rule fixes only the mean bond valence at unity. It does not force  $S_{ij} = 1$  bond by bond. For a truly degenerate shell—all bond lengths equal by symmetry, as for  $\text{Si}^{4+}$  in zircon ( $\text{ZrSiO}_4$ ), whose four Si-O bonds are symmetry-equivalent—Eq. (1) assigns every bond the same valence, and a single structure supplies only the uniform-limit relation  $R_0 = \bar{R} + B \ln(z/n)$ .  $B$  is unconstrained, and at  $z = n$  the relation collapses to  $R_0 = \bar{R}$  at any  $B$ . For a nearly degenerate shell the bond lengths differ, so the  $S_{ij}$  must differ and the sum rule is satisfied at finite  $B$ .  $\alpha$ -quartz is a common example, with two experimental

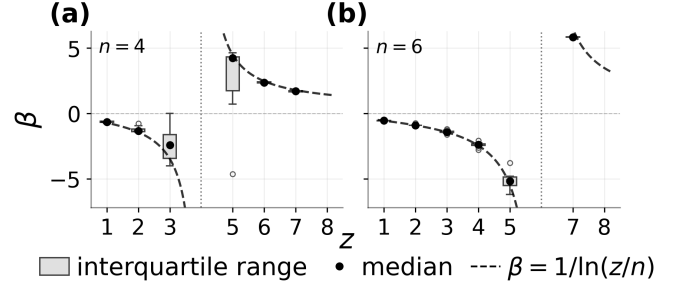


FIG. 1. Fitted slope  $\beta$  vs. formal charge  $z$  for 94 cation-oxygen species at (a)  $n = 4$  (68 cations) and (b)  $n = 6$  (82 cations). 56 cations contribute at both coordinations, giving 150 fitted slopes in total. Boxes span the interquartile range of the per-species slopes at each  $z$ , black points mark the median, whiskers extend to the most extreme slopes within  $1.5\times$  the interquartile range, and open circles mark slopes beyond the whiskers. Dashed curves:  $\beta = 1/\ln(z/n)$ . The pole at  $z = n$  separates the negative-slope ( $z < n$ ) and positive-slope ( $z > n$ ) branches of Eq. (5). Species at  $z = n$  are omitted. Per-species slopes are resolved in the Supplemental Material [11].

Si-O distances differing by about 0.01 Å that finite  $B$  resolves. Both cases are common. Of the 7,565 shells with  $z = n$  in our corpus, 959 are degenerate within  $10^{-3}$  Å and 2,960 more have bond-length spreads within 0.05 Å.

What degenerates at  $z = n$  is therefore the leading-order link between the fitted parameters. The  $B$  dependence of  $R_0$  vanishes as  $z \rightarrow n$ , so the intercept ceases to constrain the softness.  $B$  is then fixed only by the second-order spread terms  $\delta_{ij}$  in Eq. (3). The branch change across the pole is carried by the sign reversal of the slope  $\beta = 1/\ln(z/n)$ , negative for  $z < n$  and positive for  $z > n$ .

*Characteristic intersections.* Species that sit at or near  $z = n$ — $\text{B}^{3+}$  in trigonal coordination ( $\text{B}_2\text{O}_3$ , band gap 6 eV),  $\text{C}^{4+}$  at  $n = 4$ , and  $\text{Si}^{4+}$  in tetrahedral coordination ( $\text{SiO}_2$ , 9 eV)—lie in the wide-gap covalent limit where weak screening is physically plausible. For these species the pole is the limit of the first-order approximation, so the  $B$ - $R_0$  line fitted at the pole coordination cannot by itself constrain the softness. The data, however, resolve each cation at more than one coordination number, and the  $n$ -resolved  $B$ - $R_0$  lines of a given cation (slopes  $\beta^{(n)}$ ) do not scatter independently. Plotted together, they intersect at a nearly common point (Supplemental Material [11]). Because the slopes are fixed by  $z/n$ , the intersection is set entirely by the line intercepts. Writing each  $n$ -resolved line as  $B = \beta^{(n)} R_0 + \beta_0^{(n)}$ , the uniform-limit relation  $R_0 = \bar{R} + B \ln(z/n)$  gives

$$\beta_0^{(n)} = -\frac{\bar{R}^{(n)}}{\ln(z/n)}, \quad (6)$$

where  $\bar{R}^{(n)}$  is the characteristic shell radius of the species at coordination  $n$ . Each line crosses  $B = 0$  at the shell ra-

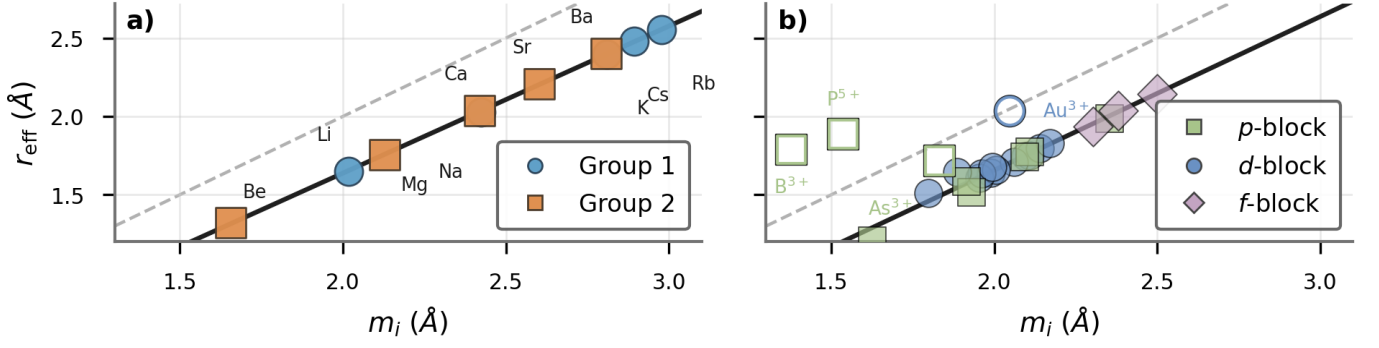


FIG. 2. First-principles screening centroid  $r_{\text{eff}}$  vs. Boltzmann shell centroid  $m_i$ . (a) Ten  $s$ -block ground-state binary oxides. The solid line is the shared fit and the dashed line is  $r_{\text{eff}} = m_i$ . (b) Twenty-five non- $s$ -block ground-state binary oxides, color-coded by block. Open symbols mark the four directional-bonding outliers excluded from the regression ( $\text{B}_2\text{O}_3$ ,  $\text{P}_2\text{O}_5$ ,  $\text{As}_2\text{O}_3$ ,  $\text{Au}_2\text{O}_3$ ).

dius itself, so the slope carries the charge-to-coordination ratio while the intercept defines the unscreened shell radius. The intersection of two coordination branches  $n_1 < n_2$  is therefore fixed by the dilation of the shell with coordination number,

$$B^* = \frac{\bar{R}^{(n_2)} - \bar{R}^{(n_1)}}{\ln(n_2/n_1)}, \quad (7)$$

with  $R_0^* = \bar{R}^{(n_1)} + B^* \ln(z/n_1)$  the common radius at which the branches agree. Bond lengths grow with coordination number, so Eq. (7) gives  $B^* > 0$  naturally on the screening branch, and the characteristic softness acquires a direct geometric meaning. It is the logarithmic rate at which the coordination shell dilates with  $n$ . These intersections define characteristic points  $(R_0^*, B^*)$ , which are species-level constants fixed by the family of lines rather than by any single member, and therefore well defined even when one line steepens toward the pole.

Within each  $(z, n)$  group, the slope  $\beta^{(n)}$  confirms Eq. (5) across all families. The  $n$ -resolved lines for a given element and oxidation state converge to characteristic pairs  $(R_0^*, B^*)$ , defined as the intersection that minimizes the weighted variance of the coordination-resolved  $B$ - $R_0$  lines, with each line weighted by the final fit population of that coordination shell (Supplemental Material [11]). Of the 103 species, 101 have  $B^* > 0$  (screening branch).

On the screening branch ( $\lambda > 0$ ,  $B > 0$ ), inverting Eq. (4) at the shell level gives the structure-specific screening length  $\lambda = (-R^* + \sqrt{R^{*2} + 4BR^*})/2$ . Evaluating the same positive-branch inversion at the characteristic point, where the operational choice of  $R^*$  equals the characteristic intercept  $R_0^*$ , gives the tabulated species constant  $\lambda^* = (-R_0^* + \sqrt{R_0^{*2} + 4B^*R_0^*})/2$ . The root is real for  $B^* \geq -R_0^*/4$ . It is positive on the screening branch and negative on the anti-screening branch.

Approaching the pole from either side ( $z = n \mp \varepsilon$ ,  $\varepsilon \rightarrow 0$ ), the leading-order relation can sustain a finite

intercept offset  $R_0 - R_0^*$  (with  $R_0^*$  the characteristic intercept introduced above) only if the fitted parameters grow as

$$|B| \sim \frac{n |R_0 - R_0^*|}{\varepsilon}, \quad |\lambda| \sim \sqrt{\frac{R^* n |R_0 - R_0^*|}{\varepsilon}}.$$

The inferred screening length grows as  $\varepsilon^{-1/2}$ . In conventional screening theory, metals have finite Thomas-Fermi screening set by a nonzero density of states at the Fermi level, whereas gapped or correlated insulators suppress low-energy charge response and therefore screen more weakly [12–15]. Our coarse-grained  $\lambda$  is not identical to those microscopic lengths, but its pole at  $z = n$  is consistent with the same weak-screening limit: the bond-valence partition becomes insensitive to finite bond-length differences and the fitted  $B$ - $R_0$  parameters cease to resolve the shell.

For the 101 screening-branch species,  $\lambda^*$  is a tabulated constant (Supplemental Material [11]). The remaining two species ( $\text{Mo}^{3+}$ ,  $\text{Sb}^{3+}$ ) have  $B^* < 0$  and fall on the anti-screening branch. The same inversion assigns them negative  $\lambda^*$  values, tabulated as signed diagnostics (Supplemental Material [11]). Both sit outside the first-order Yukawa regime:  $\text{Sb}^{3+}$  has a stereoactive  $5s^2$  lone pair driving strong directional distortion, and  $\text{Mo}^{3+}$  is constrained by only two coordination shells (Supplemental Material [11]), so its  $(R_0^*, B^*)$  is provisional. The screening-branch  $B^*$  values are non-universal (mean  $0.38 \pm 0.15$  Å), in qualitative agreement with Adams’s softBV parameterization [7], while  $\lambda^*$  provides a single-parameter species descriptor.

The second root of the inversion supplies an independent crystal-chemical check. The two roots of the quadratic behind Eq. (4) sum to  $-R_0^*$ , so on the screening branch the anti-screening alias of  $\lambda^*$  has magnitude  $|\lambda_-| = R_0^* + \lambda^*$ . This is the characteristic shell radius extended by one screening length, the outer reach of the screened bond. Compared against Shannon crystal radii

[16] averaged over each species’ observed coordination numbers,  $|\lambda_*|$  behaves as a valence-dependent contact distance (Supplemental Material [11]). Unit-slope fits  $|\lambda_*| = r_{\text{cr}} + c(z)$  hold within each formal-charge class, and the offsets follow  $c(z) \approx 1.05 + 0.09z$  Å. The constant sits at the scale of the oxygen crystal radius, so at low charge the second root reproduces the geometric cation–oxygen contact distance. The increment shows the screened bond reaching about 0.09 Å beyond that contact per unit of formal charge. The effective oxygen size seen by the bond-valence partition is therefore not a fixed sphere. It grows with cation valence, which is the statement that the bond-valence shell follows the electronic screening cloud rather than a hard ionic contact. The first-principles comparison tests this directly.

*Partition thermodynamics.* The partition of Eq. (2) carries a thermodynamic structure, developed here in three steps. First, normalize the bond valences within the shell:

$$p_{ij} := \frac{S_{ij}}{\sum_k S_{ik}} = \frac{w_{ij}}{\sum_k w_{ik}} = \frac{e^{-R_{ij}/B}}{Z_i}, \quad (8)$$

where  $Z_i := \sum_j e^{-R_{ij}/B}$ . The middle form follows from Eq. (2), whose shell total is  $z_i$ , so  $p_{ij}$  is the normalized screened-flux weight of bond  $j$ . The last form follows from Eq. (1), in which  $R_0/B$  cancels in the ratio. Equation (8) is a Gibbs distribution over the coordination shell. Bond length plays the role of energy,  $Z_i$  is the canonical partition function, and  $B$  is the temperature-like scale. Both signs of  $B$  are admissible, because a finite coordination shell has a bounded bond-length spectrum, the same condition that admits negative temperatures in bounded statistical systems. On the screening branch,  $B > 0$  acts as an ordinary positive temperature. Weight decays with bond length and the shorter bonds dominate. On the anti-screening branch,  $B < 0$  acts as a negative temperature. The shell is population-inverted, weight grows with bond length, and bonds longer than  $R_0$  carry more than one valence unit [ $S_{ij} = e^{(R_0 - R_{ij})/B} > 1$  requires  $R_{ij} > R_0$  when  $B < 0$ ]. The branches meet at the pole.  $|B| \rightarrow \infty$  is the infinite-temperature limit, where  $p_{ij} \rightarrow 1/n$  and the partition ceases to resolve bond-length differences—the screening collapse of  $z = n$ . The crossing occurs through  $1/B = 0$ , the same way a bounded system passes from positive to negative temperature. Second, define two coordination shell properties,

$$m_i := \sum_j p_{ij} R_{ij}, \quad H_i := - \sum_j p_{ij} \ln p_{ij}, \quad (9)$$

the Boltzmann mean—the valence-weighted centroid of the coordination shell—and the Shannon entropy of the bond-weight distribution. Third, invert Eq. (1) bond by bond,  $R_0 = R_{ij} + B \ln S_{ij}$ , and average over the weights  $p_{ij}$ . With  $\ln S_{ij} = \ln q_i + \ln p_{ij}$ , the average gives the exact identity

$$R_0 = m_i + B(\ln q_i - H_i), \quad (10)$$

where  $q_i := \sum_j S_{ij}$ . Here  $H_i$  measures how uniformly the bond-valence weight is distributed across the coordination shell. It reaches  $\ln n$  for a symmetric shell and falls below it as distortion concentrates weight on the extremal bonds. Thus,  $R_0$  is an entropy-corrected weighted mean. It reduces to the unit-valence limit  $R_0 = R_{ij}$  that Brown *et al.* [4] derived for a single bond with  $S_{ij} = 1$ . The Boltzmann mean also fixes the reference length left open in the screened-flux derivation. We take  $R^* = m_i$ , which centers the log-weight expansion of Eq. (3) on the weighted shell centroid that the partition function singles out and makes the operational content of  $R^*$  in Eq. (4) explicit.

*Empirical validation.* Bond-valence parameters  $(R_0, B)$  were determined for 103 cation–oxygen species using first-principles relaxed crystal structures from the Materials Project [17, 18]. For each structure, bond strengths were obtained by solving the valence-sum and loop constraints on the cation–oxygen bond graph [19], and  $(R_0, B)$  were fitted by linear least squares. Each structure thus contributes one  $(R_0, B)$  pair with both parameters fitted jointly, rather than  $B$  being re-optimized at fixed  $R_0$ . The coordination-resolved lines of Fig. 1 are then obtained by regressing  $B$  on  $R_0$  across the per-structure pairs of each species at fixed  $n$  (at least five structures per cell), using an outlier-robust RANSAC estimator refit by least squares on its inliers. A fitted slope  $\beta^{(n)}$  is accepted only with at least four inlier structures, slope uncertainty  $\sigma_\beta \leq 5$ , and  $R^2 \geq 0.3$  [11, 20]. The resulting set spans  $s$ -,  $p$ -,  $d$ -, and  $f$ -block cations. Figure 1 shows the 94 species (150 fitted slopes) for which the prediction is finite at  $n = 4$  and/or  $n = 6$ .

*First-principles confirmation.* The connection to electronic structure is tested by comparing the bond-valence Boltzmann shell centroid  $m_i$  of Eq. (9) to a first-principles screening centroid  $r_{\text{eff}}$  extracted from charge densities in the Materials Project [17, 18]. For each M–O bond with  $R_{ij} \leq 3.5$  Å, the total valence-electron density  $\rho(r)$  is sampled at 200 uniformly spaced points along the nearest-image straight M–O segment. An interior density-minimum proxy on  $0.15R_{ij} < r < 0.85R_{ij}$  defines the start of the oxygen-side window, and  $r_{\text{eff}}$  is then the excess-density-weighted centroid on  $r_{\text{BCP}} \leq r \leq 0.93R_{ij}$  with weights  $\max[\rho(r) - \rho_{\text{BCP}}, 0]$ , averaged over the valid first-shell bonds of that oxide (see Supplemental Material for the exact discrete construction and fallback rule). For ten  $s$ -block binary oxides (Li<sub>2</sub>O through BaO, Fig. 2a), these collapse onto a single line:

$$r_{\text{eff}} = 0.94 m_i - 0.24 \text{ Å} \quad (R^2 = 0.9998).$$

This is not a trivial monotone size trend: the jackknife mean absolute error of the  $m_i$  regression is 0.006 Å, versus 0.078 Å for a linear fit against  $R_0$  and 0.095 Å against the Shannon cation radius [16] (see Supplemental Material for the corresponding control plot). The near-unit slope therefore shows that the bond-valence



partition function and the first-principles screening cloud track the same ionic shell radius. The same centroid correspondence extends beyond the  $s$  block. Figure 2b shows the 25 non- $s$ -block binary oxides for which the charge density yields a well-defined oxide-level screening centroid (listed in the Supplemental Material [11]). The Yukawa derivation assumes approximately isotropic first-shell screening, so the regression is restricted to the 21 oxides whose cation environments satisfy that condition, excluding four whose bonding is dominated by directional, non-isotropic motifs:  $\text{B}_2\text{O}_3$ ,  $\text{P}_2\text{O}_5$ ,  $\text{As}_2\text{O}_3$ , and  $\text{Au}_2\text{O}_3$ . The Supplemental Material gives the fixed inclusion rule and the exact oxide-level centroid construction used to define  $r_{\text{eff}}$ . The 21 included oxides follow

$$r_{\text{eff}} = 0.98 m_i - 0.30 \text{ \AA} \quad (R^2 = 0.967).$$

The correspondence therefore spans the  $p$ ,  $d$ , and  $f$  blocks, and its breakdown is confined to strongly anisotropic directional-bonding environments outside the isotropic Yukawa regime.

*Discussion.* The main result is that the bond-valence exponential is the leading-order term of Yukawa screened electrostatics. The fitted exponent denominator  $B$  is not a free parameter but the leading-order Yukawa denominator set by the screening length and shell geometry, with its sign carrying the screening/anti-screening branch information. The empirical correlation between  $R_0$  and  $B$  is determined by  $\beta = 1/\ln(z/n)$  and exact in the uniform-bond limit. The partition-function structure for inequivalent bonds gives an entropy-corrected identity for  $R_0$  and a free energy  $F = m_i - BH_i = R_0 - B \ln q_i$  in length units.

The derivation rests on two approximations: (i) the first-order Taylor expansion, which breaks down for highly distorted shells ( $d^4/d^9$  Jahn–Teller octahedra), and (ii) isotropic flux capture, in which the ansatz  $w_{ij} \propto e^{-R_{ij}/\lambda}(1 + R_{ij}/\lambda)$  assigns each bond a weight that depends on its length alone.

as in stereoactive lone-pair and strongly anisotropic environments [4, 21]. The empirical signature of this breakdown is visible in Fig. 2b: the four oxides whose ground-state structures fail the isotropic flux-capture assumption ( $\text{B}_2\text{O}_3$ ,  $\text{P}_2\text{O}_5$ ,  $\text{As}_2\text{O}_3$ ,  $\text{Au}_2\text{O}_3$ ) are precisely those that depart from the centroid line, identifying ground-state directional bonding as the empirical boundary of the first-order Yukawa regime. Extending the first-principles centroid comparison to the full set of transition-metal and  $f$ -block oxides beyond the subset treated here will require going beyond the Thomas–Fermi approximation to include orbital directionality and correlation effects.

The first-principles centroid comparison confirms that this is not merely a mathematical analogy, but that the electronic screening cloud and the bond-valence weights describe the same shell geometry. The small deviation from the identity line is explained by the oxygen-side valence centroid shift measured directly from the same

charge-density paths, while the null-model controls show that the collapse is substantially sharper than a generic monotone size trend.

Bond valence is therefore recast as a coarse-grained observable of local screened charge response rather than a purely empirical structural heuristic. Concretely, the species-specific screening length  $\lambda^*$  is a compact tabulated descriptor that organizes the shell thermodynamics and, through the identities above, provides a calibrated starting point for estimating bond weights, entropy, effective coordination number, and  $R_0$  in new coordination environments without additional adjustable parameters. Establishing the full predictive scope of that program beyond the families validated here remains future work. This connects Pauling’s 1929 valence rules [1] to Yukawa’s 1935 screened potential [22] through a quantitative bridge that has been missing for nearly a century.

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